

Introduction to statistics

Correlation and regression

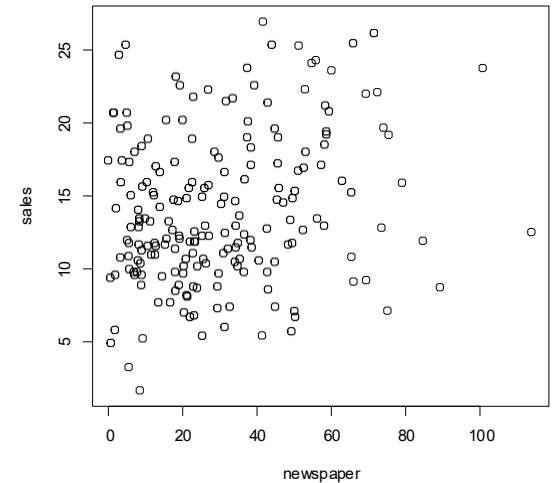
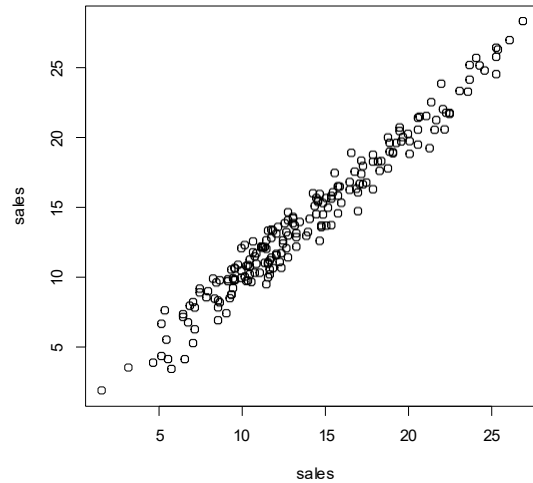
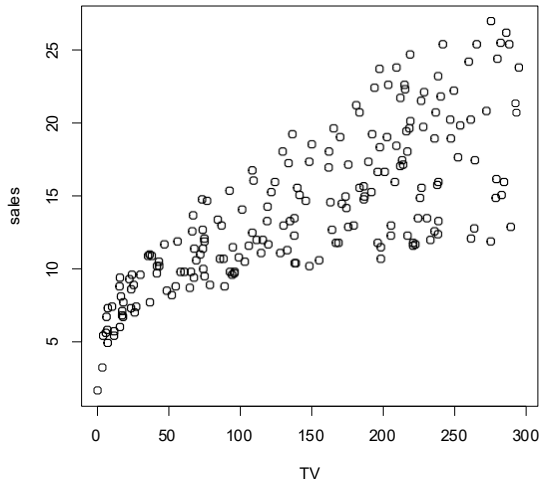
Tallinn, June 2026

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(SIB)

Modified by Marilin Moor (UT)



Scatterplot



We are often interested in the statistical dependence between two variables, aka “correlation”

Pearson correlation

- Is a measure of linear association
- Pearson correlation coefficient (r – sample, ρ – distribution) indicates the strength of a linear relationship between two variables (uses covariance and standard deviation)

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient

x_i = values of the x-variable in a sample

$\bar{x}$ = mean of the values of the x-variable

y_i = values of the y-variable in a sample

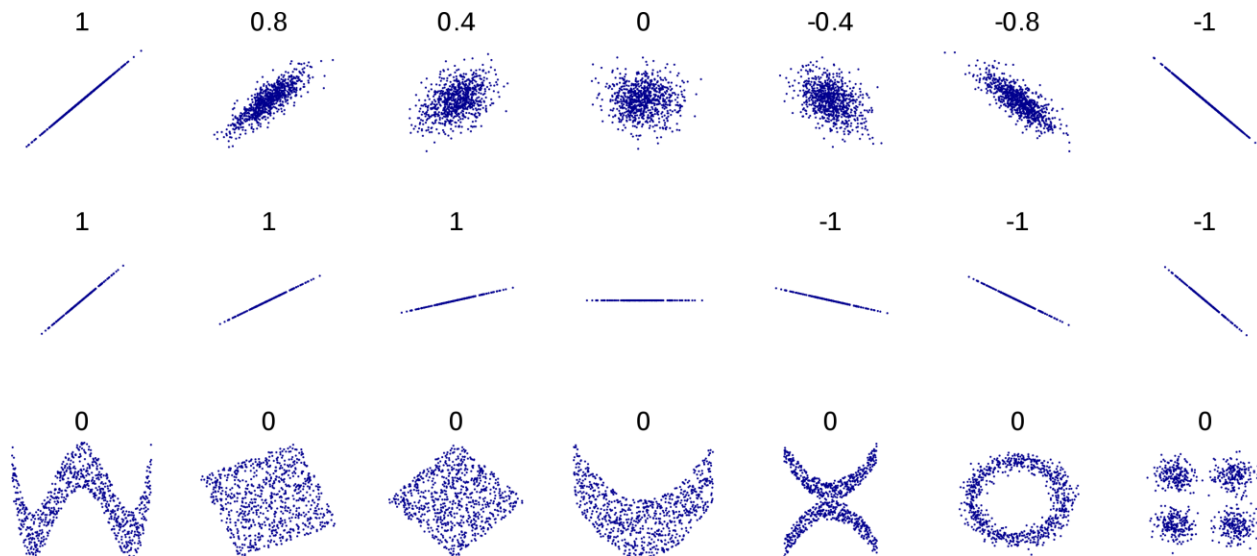
$\bar{y}$ = mean of the values of the y-variable

Pearson correlation

$$-1 \leq r \leq 1$$

r is a *unit-less quantity*

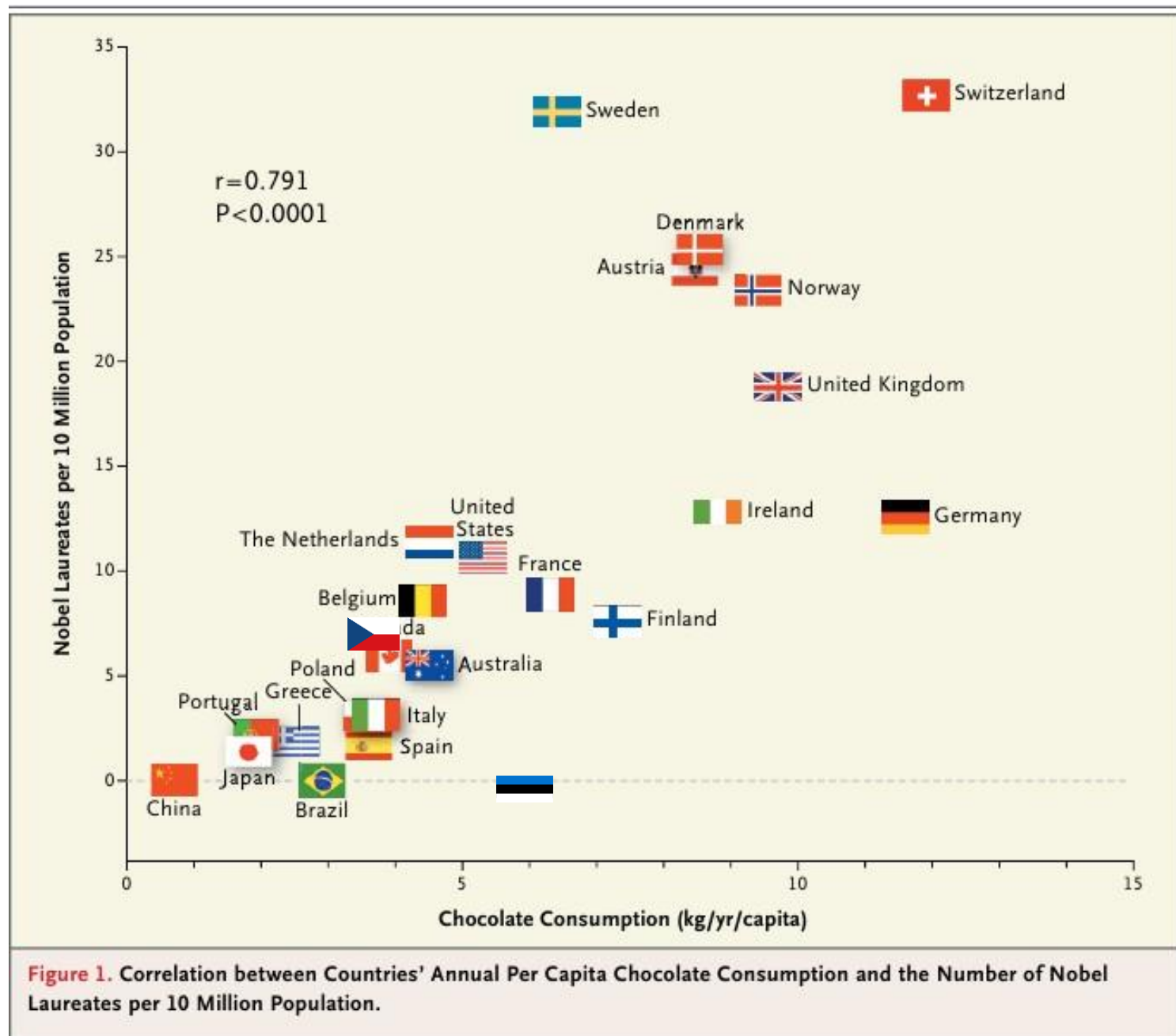
the closer r is to -1 or 1 , the more tightly the points on the scatterplot are clustered around a line



https://upload.wikimedia.org/wikipedia/commons/thumb/d/d4/Correlation_examples2.svg/1920px-Correlation_examples2.svg.png

CORRELATION IS NOT CAUSATION

- You *cannot* infer that since X and Y are highly correlated (r close to -1 or 1), X is *causing* a change in Y
- Y could be causing X
- X and Y could both be varying along with a third, possibly unknown variable (either causal or not)



tylervigen.com

Spurious correlations

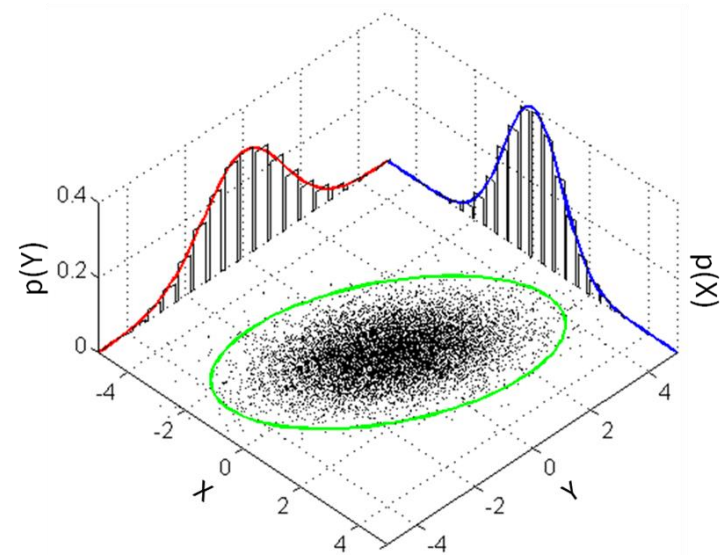


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Assumptions of Pearson correlation

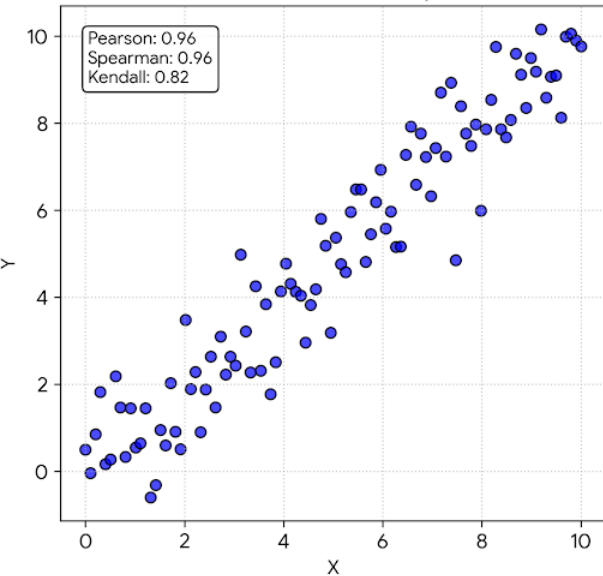
- The only assumption of Pearson correlation is that the data follows a bivariate normal distribution



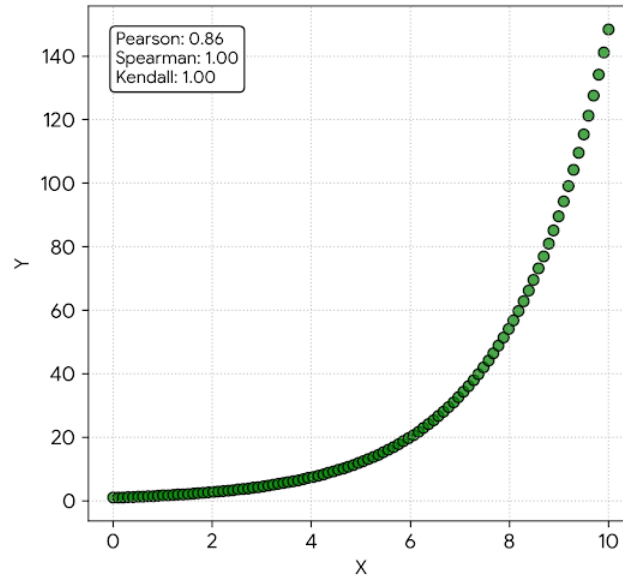
- When this assumption is not met, alternative measures of association between two variables should be used
 - Spearman rank correlation (considers monotonicity/order, better with outliers)
 - Kendal rank correlation (similar to Spearman but for smaller dataset and identical data values)

Correlation comparison

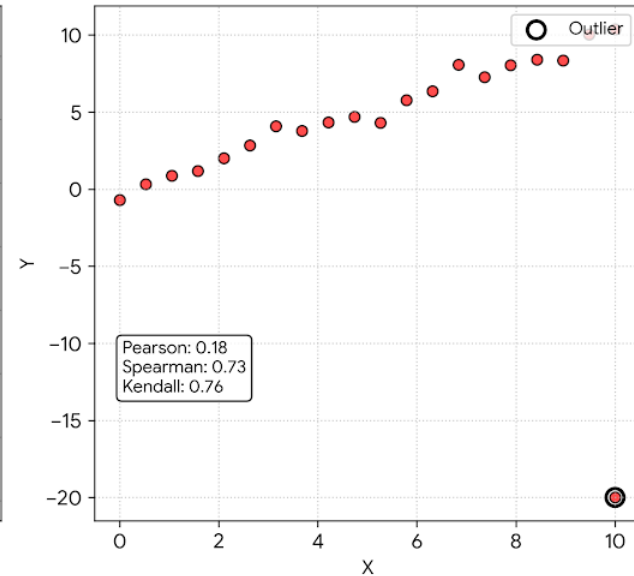
Linear Relationship



Monotonic Non-Linear



Linear with One Severe Outlier



```
x <- c(35, 20, 49, 44, 30)
y <- c(24, 35, 39, 48, 45)
cor.test(x,y)
```

Pearson's product-moment correlation

```
data:  x and y
t = 0.47157, df = 3, p-value = 0.6694
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
-0.8064873  0.9295282
sample estimates:
      cor
0.2626993
```

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

NB! The p-value measures how likely you would be to observe a correlation of this strength under the null hypothesis.

While H0 is that true correlation is 0, and alternative is that it is not 0. Here the sample size is so small that t-statistic falls into the H0 region.

```
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What to report? number of data points (>30 sample size needed), the correlation coefficient (two decimal places), together with the p-value and the 95% confidence interval

```
> cor.test(x,y, method="spearman")
```

Spearman's rank correlation rho

data: x and y

S = 14, p-value = 0.6833

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

0.3

```
> cor.test(x,y, method="kendall")
```

Kendall's rank correlation tau

data: x and y

T = 6, p-value = 0.8167

alternative hypothesis: true tau is not equal to 0

sample estimates:

tau

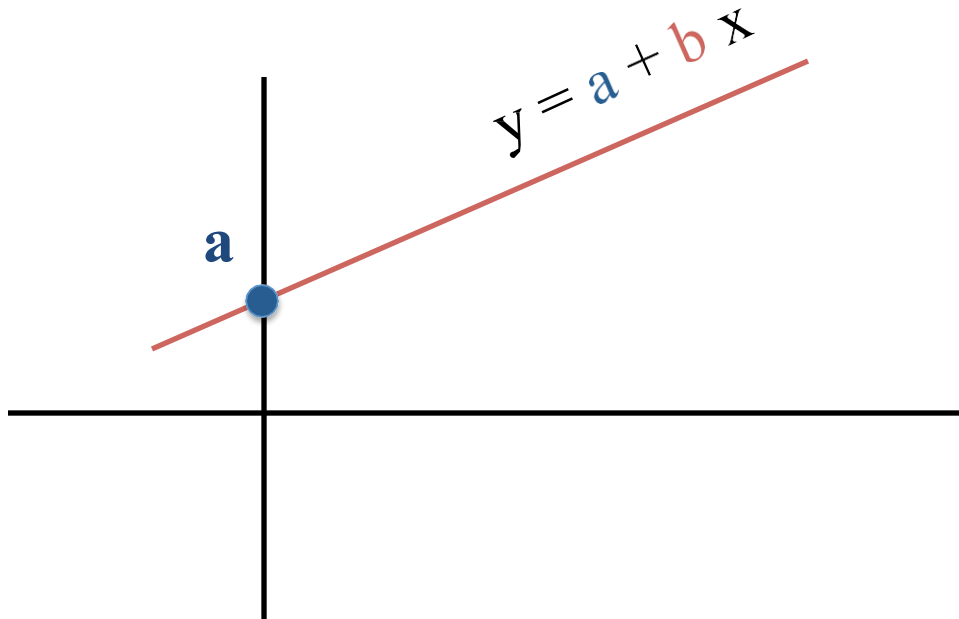
0.2

- **Correlation** describes the association between variables, but does not describe it
- Often it is useful to obtain a mathematical model that describes the association between variables, hence **regression**

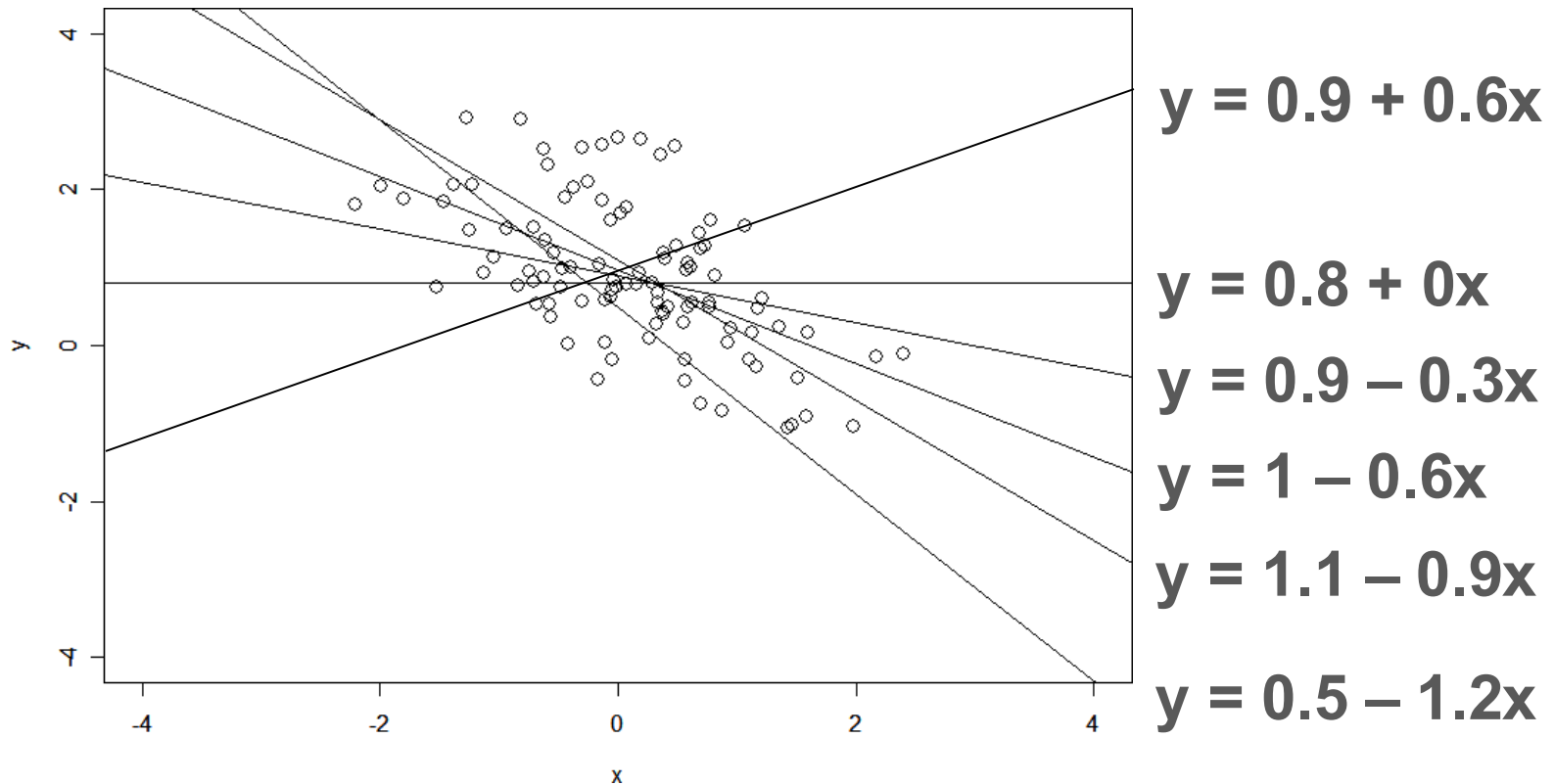
The equation for a line that can be used to predict y knowing x (in slope-intercept form) looks like

$$y = a + b \cdot x$$

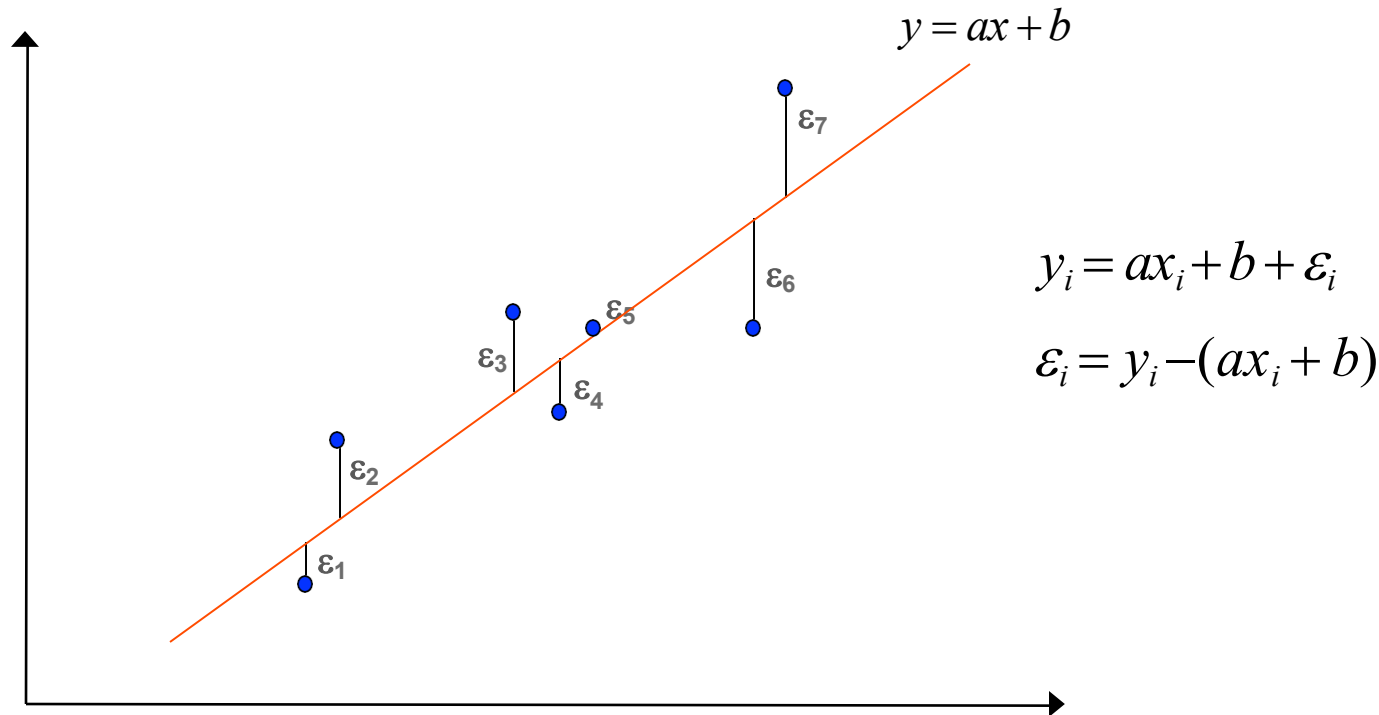
where a is called the *intercept* and b is the *slope*.



What is the “best” line that fits this data ? → need a criteria
Can we use it to summarize the relation between x and y ?



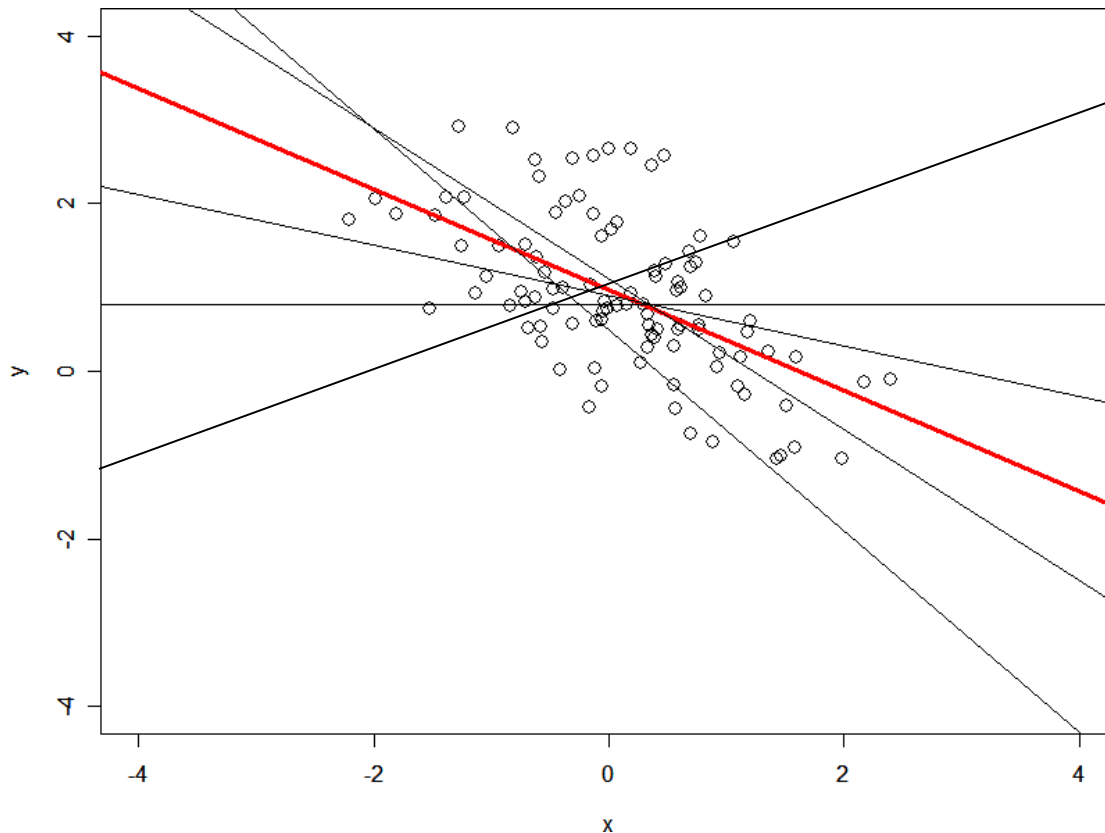
Least-squares approach to fit a line



The least-squares procedure finds the straight line with the **smallest sum of squares of vertical errors**.

Finds a regression line such that $\sum_i \varepsilon_i^2 = \varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \dots$ is minimum.

Over all possible straight lines,
 $y = 1 - 0.6x$ is the “best” possible line
according to least-squares criterion



$$y = 0.9 + 0.6x$$

$$y = 0.8 + 0x$$

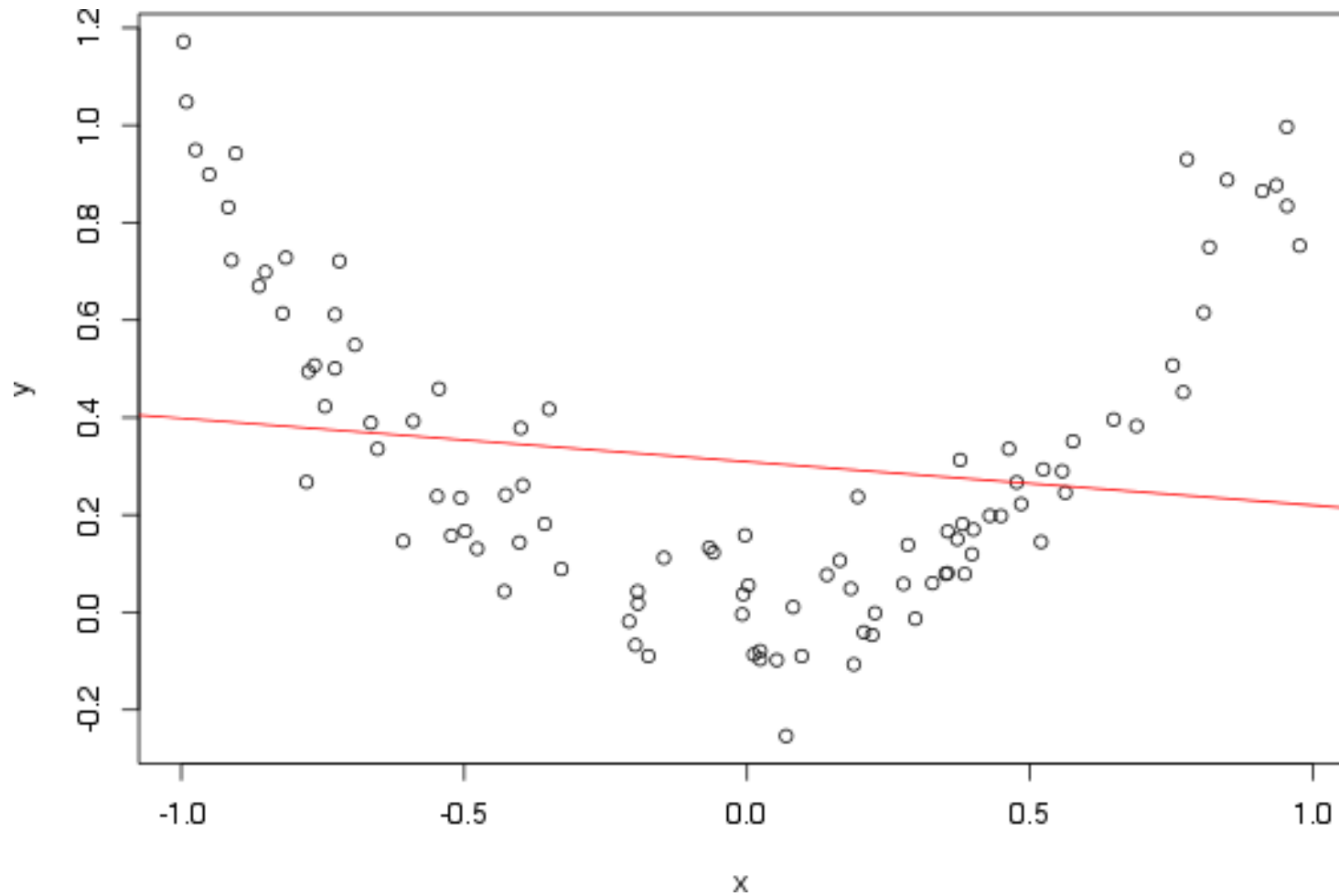
$$y = 0.9 - 0.3x$$

$$y = 1 - 0.6x$$

$$y = 1.1 - 0.9x$$

$$y = 0.5 - 1.2x$$

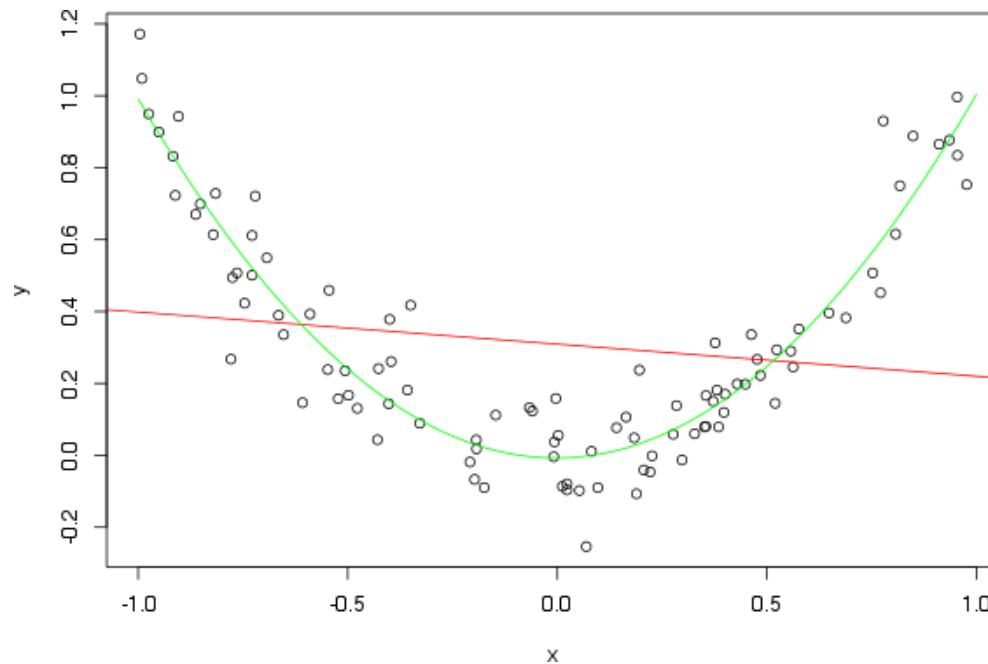
What if the association is not linear ?



What if the data is not linear ?

Use a polynomial regression

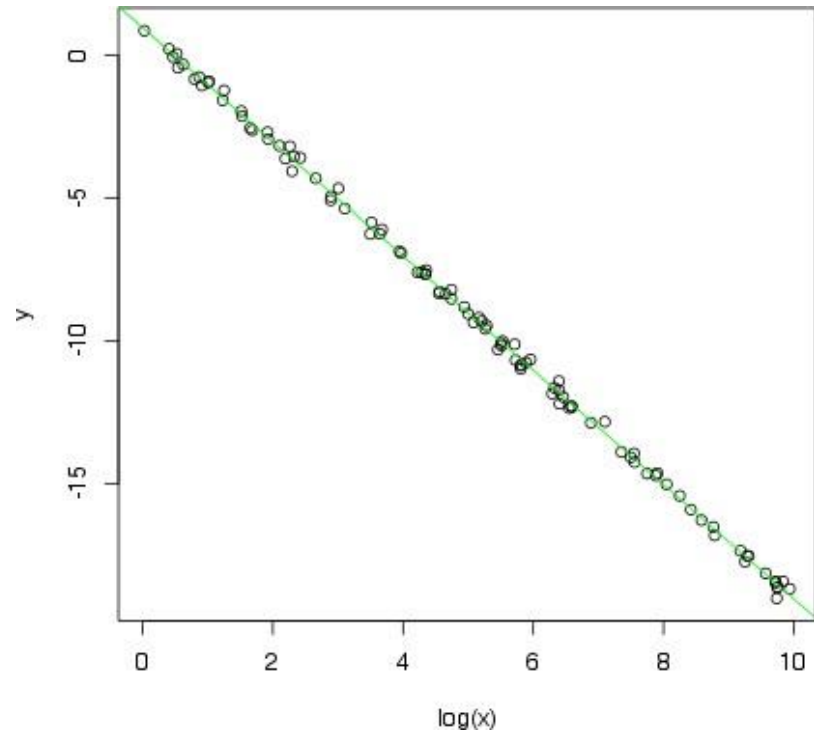
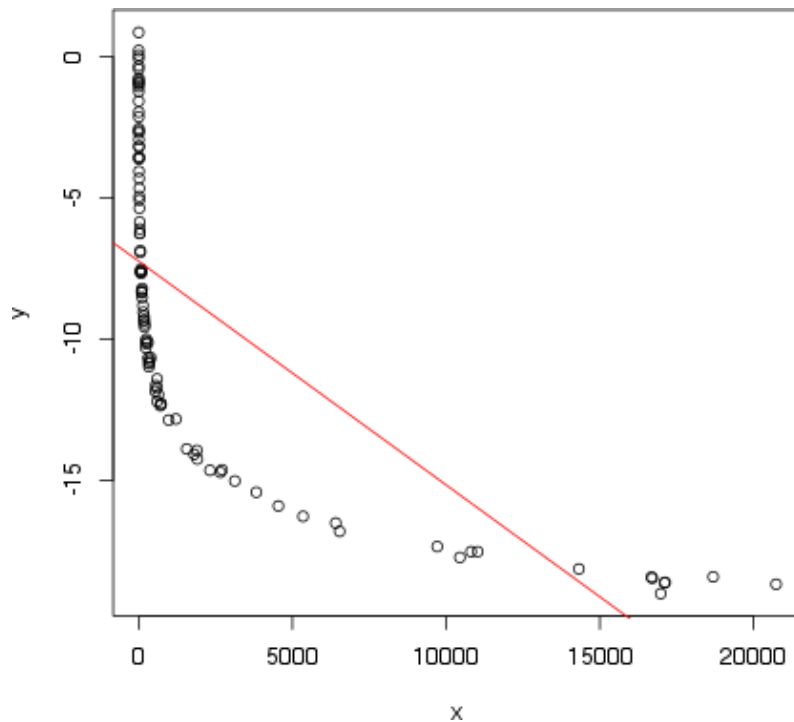
$$y = b_0 + b_1 x + b_2 x^2$$



What if the association is not linear ?

Consider transforming the data (log)

$$\log(y) = a + bx$$



$$Y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3$$

in R:

```
lm(yvar ~ xvar1 + xvar2 + xvar3)
```

read “~” as “described (or modeled) by”

By default, an intercept is included in the model

To leave the intercept out:

```
yvar ~ -1 + xvar1 + xvar2 + xvar3  
yvar ~ 0 + xvar1 + xvar2 + xvar3
```